SALAR TAHERI

Senior Android & Mobile Systems Engineer

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Professional Summary

Senior Android & Mobile Systems Engineer with over **10 years of production experience** specializing in Fintech architectures, Android POS hardware integration, low-level systems communication, and high-scale enterprise ecosystems. Track record architecting mission-critical biometric eKYC pipelines serving **2.5M+ active users**, payment SDKs processing **100k+ daily transactions**, and real-time transaction processing strictly compliant with central banking standards (ISO 8583, Shaparak Kehroba contactless NFC – EMV / Apple Pay equivalent protocol). Deep expertise spanning modern Android engineering (Jetpack Compose, Kotlin Coroutines & Flow, Clean Architecture, DDD) and low-level peripheral systems (C/C++ via JNI/NDK, AIDL, serial/RS232/USB SPP protocols, APDU, Java Card applets). Proven technical leader establishing CI/CD automation, setting code review standards, mentoring engineers, and leveraging AI-assisted engineering workflows (Cursor, Copilot, Agentic Tooling) for automated test scaffolding and boilerplate elimination.

10+ Years	2.5M+ Users	<200ms Latency	99.8%
Engineering Experience	Biometric eKYC SDK	ISO 8583 Socket Clearance	Crash-Free Rate

Technical Skills

- **Languages:** Kotlin, Java, C/C++ (JNI / NDK), Dart, SQL, Bash
- **Android Core & Systems:** Android SDK, Jetpack Compose, View System & 2D Canvas, AIDL, Host Card Emulation (HCE), CameraX, Navigation Component, Material Design 3, Linux Kernel Tuning, In-Memory Bitmap Rendering
- **Architecture & Concurrency:** Clean Architecture, Domain-Driven Design (DDD), MVVM, Multi-Module Gradle, Kotlin Coroutines, Kotlin Flow, RxJava, Asynchronous Thread Pooling, Memory & GC Pause Optimization
- **Fintech & Hardware Protocols:** ISO 8583, JPOS, Shaparak Kehroba (NFC Contactless / EMV Equivalent), Java Card (Applets), APDU Protocol, Pax & Amp POS SDKs, ESC/POS Thermal Printers, Serial / RS232 / USB / Bluetooth SPP, Smart Card Readers, PIN-pad Integration
- **Networking & Data:** Asynchronous TCP Sockets, WebSockets, Apache ActiveMQ, MQTT, STOMP, Ktor Client, Retrofit, OkHttp, REST APIs, Protobuf, Room Database, SQLite, DataStore, Hilt, Koin
- **Testing, QA & Mocking:** JUnit 5, MockK, Robolectric, Turbine, Hardware-in-the-Loop (HIL) Emulation, Mock TCP & APDU Sockets, Simulated Peripheral Drivers, End-to-End Payment Integration Testing
- **Security, DevOps & AI Tools:** Android Keystore, Anti-Tamper / Root Detection, ProGuard / R8 Obfuscation, CI/CD (GitHub Actions), Docker, Git, Linux, Sentry, AI-Assisted Development (Cursor, GitHub Copilot, Agentic Tooling), Prompt Engineering for Automated Testing & Scaffolding

Professional Experience

Senior Android Developer
Mar 2023 – Present

AvaParsi

- **Offline-First POS Architecture & Memory Tuning:** Architected and scaled offline-first Android POS app for supermarkets and restaurants; utilized Room Persistence, multi-tier in-memory caching, and GC optimization to handle 50k+ product catalogs with zero-latency lookups.
- **Cryptographic Tax Compliance System:** Engineered the AmnPardaz Electronic Invoicing System in Flutter/Android, enabling corporate merchants to submit cryptographically signed tax invoices to the national tax authority via smart POS terminals (**50k+ invoices**, 100% acceptance).
- **Hardware Integration & AIDL:** Designed queue management and ticketing workflows by interfacing directly with digital weight scales, barcode scanners, and ESC/POS thermal receipt printers via AIDL, USB SPP, and RS232 serial protocols.
- **Hardware-in-the-Loop (HIL) Emulation & Mock Sockets:** Containerized mock APDU/TCP server environments and simulated serial drivers using Docker to test peripheral responses and transaction sequences in CI pipelines without physical POS hardware, expanding automated test coverage by 40%.
- **DDD Architecture & Cross-Vendor Stability:** Led architectural refactoring toward Domain-Driven Design (DDD) and modular multi-module Gradle packaging, sustaining a **99.8% crash-free rate** across fragmented Android POS device vendors (Pax, Amp, Newpos, Centerm).

Android Systems Consultant (Part-Time / Advisory)
Oct 2023 – Present

GSS (GSS International)

- **GMS-Free Sovereign POS Supervision:** Architected Google-independent real-time supervision and alert service for Android POS devices utilizing a custom STOMP/MQTT client over Apache ActiveMQ, eliminating Google FCM dependencies across banking networks.
- **Resilient Background Architecture & Power Tuning:** Engineered an always-on Foreground Service with Partial WakeLocks and exponential backoff auto-reconnect, sustaining **99.9% socket uptime** with minimal battery and cellular bandwidth consumption.
- **Nationwide Telemetry & Bi-Directional File Updates:** Streamed hardware diagnostics (battery, thermal, 4G RSSI, peripherals) and executed chunked file transfers (crash logs, OTA silent APKs) across **10,000+ Android POS terminals** with sub-second alert delivery.

Android Developer (Fintech & POS) *Sep 2022 – Mar 2023*

Omidpay

- **Low-Level Banking Transaction SDKs:** Engineered low-level banking SDKs for Android smart POS terminals (Pax A920Pro, Amp8000), implementing direct communication layers for magnetic stripe cards, EMV smart IC cards, and hardware-secured PIN-pads.
- **ISO 8583 Protocol & Sub-200ms Clearance:** Implemented strict ISO 8583 message encoders/decoders, JPOS standards, and asynchronous TCP socket streaming, optimizing network buffers and thread pools to achieve **sub-200ms latency** for interbank transaction clearance.
- **Nanino Smart Bakery Platform:** Shipped the Android POS client for the Nanino platform – a nationwide smart bakery retail transaction infrastructure – powering thousands of merchants to process **100k+ daily transactions** with zero data loss.
- **Automated Socket & APDU Mock Testing:** Built automated mock TCP socket suites and APDU response emulators with JUnit 5 and Robolectric to validate ISO financial messaging and edge-case network dropouts in CI pipelines without physical POS hardware.
- **Cybersecurity Hardening & Certification:** Hardened payment binaries with anti-hooking/anti-tamper detection, native C/C++ checks, and Android Keystore encryption, successfully passing rigorous banking security audits by the National Cybersecurity Certification Body (AFTA).
- **OEM & Cross-Functional Collaboration:** Partnered closely with hardware OEMs (Pax, Amp) to resolve low-level firmware quirks and collaborated with backend architects to streamline financial switch payloads.

Android Developer (eKYC Platform) *Sep 2018 – Sep 2022*

UID

- **National-Scale Biometric eKYC SDK:** Engineered the core client SDK for Iran's premier digital identity verification platform, enabling automated liveness detection, AI-driven facial verification, and real-time video streaming over WebSockets for **2.5M+ active users** across major banks and national capital market onboarding (Sejam).
- **Architecture Modernization & Performance:** Spearheaded legacy codebase refactoring from Java/MVP to Kotlin/MVVM and Coroutines, reducing APK footprint by **35%**, optimizing memory allocation, and drastically cutting external runtime dependencies.
- **Biometric Security & Obfuscation:** Configured custom ProGuard/R8 bytecode obfuscation rules, anti-hooking detection, and Android Keystore payload encryption to protect biometric telemetry in transit and at rest.
- **Developer Integration Experience:** Authored technical integration documentation, client SDK sample implementations, and sandbox test harnesses, accelerating enterprise integration across 30+ financial institutions.

Android Developer *2016 – 2018*

Freelance (Softwaria / Varna Marlik)

- **Municipal Fleet Dispatch & Printing Engine:** Built the Tonekabon Municipal Taxi System featuring an in-memory 2D Canvas rendering engine streaming dynamic receipts as bitmaps over Bluetooth/Serial SPP to Bixolon thermal printers (<**800ms** print latency, 100% field adoption).
- **Native 2D Canvas Game Engine:** Designed and shipped *Dolme* – a native Persian puzzle game on CafeBazaar (regional Android marketplace with 40M+ active users) with 60 FPS gameplay via custom Canvas Views and animation loops without third-party game engines (**50k+ downloads, 4.7/5 rating**).
- **Commercial Mobile Solutions:** Built and deployed on-demand food ordering, classifieds, and developer utility applications for regional commercial clients.

Key Projects

Android POS Banking & Kehroba Contactless System *Sep 2024 – Mar 2025*

Independently architected and implemented a complete banking application for smart POS terminals. Implemented ISO 8583 binary protocol over raw TCP sockets, Java Card APDU smart card authorization, and contactless NFC transactions via Host Card Emulation (HCE) compliant with Shaparak Kehroba (*National Central Bank contactless payment scheme, EMV / Apple Pay equivalent protocol*). Achieved **sub-200ms transaction clearance latency** and **100% central banking compliance**.

LinuxNetwork.ir – Linux Network & Kernel Tuning Toolbox *2025*

High-performance bilingual web engineering toolbox (React, TypeScript, Tailwind CSS) for Linux kernel tuning (`sysctl`, BBR TCP congestion control), WireGuard VPN configuration, and CIDR subnet calculation. **100/100 Lighthouse score**, sub-100ms edge delivery via Cloudflare Pages, thousands of monthly DevOps visitors.

Vira – Smart Document Image Processing Service *2025*

Intelligent client-side document photo normalization platform built with HTML5 Canvas and Web Workers. Processed **20,000+ applicant photos** for educational institutions, reducing rejection rate from 18% to <**0.2%** (<50ms in-browser latency, zero server upload privacy guarantee).

Education

Bachelor of Science (B.Sc.) in Software Engineering *2013 – 2018*

University of Guilan, Iran

Certifications & Compliance

- **National Cybersecurity Compliance (AFTA):** Certified secure payment application development, cryptographic key management, and anti-tamper hardening for banking POS terminals.
- **Central Banking Payment Standards:** Verified implementation of Shaparak Kehroba contactless NFC payment protocol and ISO 8583 banking switch integrations.

Languages

Persian – Native — **English** – Professional Working Proficiency (Technical Documentation, Team Collaboration)